Everyday memory support, built around people

A cognitive engagement and memory assistance platform for older adults and the people who care for them.

Why we started

Care does not happen only during a hospital visit. An older person may need help beginning an activity, recognising a familiar face, or keeping track of a routine. Families often provide that support while trying to understand how the day has gone.

We started SMĀRANA with those everyday moments in mind. We also wanted to consider the realities of the North East, where language, digital familiarity and unreliable connectivity can affect whether a tool is useful.

Our idea

SMĀRANA brings short cognitive activities, familiar personal cues and routine support into one accessible experience. An assigned caregiver can see shared activity when it has synchronised. We want the experience to be simple enough to use at a comfortable pace, with room for people to stop and return later.

The platform is intended for older adults living with mild cognitive impairment or mild-to-moderate dementia. It is an assistive tool. It does not diagnose dementia, make treatment decisions or replace a clinician.

What a person and caregiver can do

Engage through activity. Short activities such as memory matching offer a manageable way to take part. The prototype shows a memory game and a rules-based approach to adjusting difficulty.

Use familiar cues. The wider concept includes names, photographs and familiar moments supplied with consent. These can help make the experience personal and recognisable.

Support routines. The proposed experience includes cues for everyday activities and appointments. A confirmation records what the person reports; it does not prove that medication was taken.

Share context. An assigned caregiver can review synchronised activity. We have a caregiver screen and have run an isolated readback test using synthetic records.

How the system works

The patient-facing app is a React and TypeScript progressive web app. Activity can be stored on the device using IndexedDB and Dexie. An outbox holds queued events while the device is offline. When the app is active or reopened and connectivity returns, events can be sent through an authenticated Django REST API. PostgreSQL stores shared application records.

In an isolated browser run, we demonstrated local save, synchronisation and caregiver readback. It does not establish closed-app background sync or acceptance on physical phones. We need to test those conditions before relying on them.

Adapting the experience

The current design uses clear rules to adjust game difficulty. After a round, the system can consider accuracy, response time and hints before making a small change to the next activity. These signals can be affected by tiredness, distraction, motor ability and familiarity with the device. They should never be treated as a diagnosis or a measure of disease progression.

Language and voice

The prototype shows a language selector and an Assamese interface. We want to explore voice input and spoken responses in regional languages, with touch controls available as an alternative. Voice is part of the broader vision and still needs development and testing with native speakers, older users and the devices they use.

Where the project stands

We have browser captures of a daily session, a memory game and a caregiver screen. The caregiver view comes from a separate run, and the records shown are synthetic. We have also run an isolated offline save and readback test. We have not established clinical benefit or readiness for field deployment.

Before a pilot, we need to verify the current build, review the activity content and difficulty rules, test language and accessibility, and check the reliability of synchronisation and reminders on target devices. Alerts should not be presented as an emergency service without a tested delivery and escalation process.

Privacy, safety and consent

People should be able to decide what personal information is used and who can see it. We intend to limit access to assigned accounts, collect only what is needed, and provide ways to correct or remove records. Encryption, recovery, retention and access controls need to be verified before field use.

A clinical advisor should review activities and safety wording. The app should not change prescriptions or give medical advice, and important care decisions should remain with people and their clinicians.

What we want to do next

Our next step is to work with a clinical mentor, native-language and accessibility reviewers, and an elder-care or healthcare partner to design a small pilot. We would measure whether people can complete representative tasks, whether they return to activities, and whether caregivers find the shared context useful.

We have outlined a larger 12-month pilot as a planning proposal. Its scope, budget, approvals and partner commitments would need to be agreed before it could begin. Our first goal is to learn what works, make the product safer and easier to use, and decide what is worth expanding.